

Course Project AO:

Mathematical Analysis of Algebraic-Optimized (AO) Ciphers

Introduction and Motivation

Traditional block ciphers such as AES are designed primarily for efficient software and hardware implementation while providing strong resistance against classical cryptanalytic techniques. In contrast, a growing class of modern cryptographic primitives is designed with a different optimization goal: *algebraic efficiency*.

These *algebraic-optimized (AO) ciphers* aim to minimize quantities such as:

- the number of nonlinear (AND or multiplication) operations,
- the algebraic degree of intermediate computations,
- the multiplicative depth of the corresponding Boolean or arithmetic circuit.

Such designs are motivated by applications in secure multi-party computation (MPC), fully homomorphic encryption (FHE), and zero-knowledge proof systems, where the cost of evaluating nonlinear operations dominates performance. As a result, AO ciphers trade traditional implementation efficiency for algebraic simplicity.

This project explores AO ciphers from a *mathematical perspective*, focusing on their Boolean structure, algebraic degree growth, linear diffusion layers, and polynomial representations.

AO Ciphers in the Literature

We briefly describe several prominent AO ciphers and permutations that will serve as concrete objects of study in this project.

LowMC

LowMC is a block cipher explicitly designed to minimize the number of AND gates in its Boolean circuit representation. Each round consists of:

- a small number of low-degree S-boxes (typically quadratic),
- a linear diffusion layer over $\mathbb{F}_2$,
- round key addition.

LowMC deliberately delays algebraic degree growth across rounds, making it well-suited for MPC and FHE, but potentially more vulnerable to algebraic analysis.

FLIP

FLIP is an extreme example of an AO cipher, designed with an exceptionally low multiplicative complexity. Its nonlinear layer is deliberately simple, which makes it an excellent object for studying algebraic degree growth and the limits of algebraic cryptanalysis.

FASTA and Related Designs

FASTA and related constructions aim to reduce multiplicative depth even further than LowMC, often using carefully structured linear layers and minimal nonlinear components. These designs appear frequently in recent work on ZK-friendly cryptographic primitives.

AIM (Algebraic Intermediate Model) and RAIN

AIM: Symmetric Primitive for Shorter Signatures with Stronger Security (<https://eprint.iacr.org/2022/1387>)

AO Permutations

While not block ciphers in the traditional sense, algebraic-optimized permutations such as Poseidon, Rescue-Prime, Griffin, and Anemoi dominate modern zero-knowledge systems. Many of their design principles mirror those of AO block ciphers and provide valuable comparative insight.

Project Structure

This project consists of **three independent mathematical tracks**. Each student (or group) can choose **one** of the tracks described below. All tracks require careful mathematical reasoning, formal definitions, and precise argumentation.

1 Track 1: Algebraic Normal Form and Degree Growth

Objective

The goal of this track is to analyze the Boolean structure of an AO cipher by expressing its round function in *algebraic normal form (ANF)* and studying the growth of algebraic degree across rounds.

Background

Any Boolean function

$$f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$$

can be uniquely represented as a polynomial over $\mathbb{F}_2$, known as its algebraic normal form:

$$f(x_1, \dots, x_n) = \bigoplus_{S \subseteq \{1, \dots, n\}} a_S \prod_{i \in S} x_i.$$

The *algebraic degree* of f is the maximum size of a set S such that $a_S \neq 0$.

Project Description

Choose an AO cipher (e.g., LowMC or FLIP) with a small block size. Derive the ANF representation of:

- the nonlinear layer of one round,
- the full round function,
- the composition of multiple rounds.

Using these representations, analyze how the algebraic degree of output bits evolves as the number of rounds increases. Compare this growth to known bounds for random Boolean functions and discuss why AO ciphers deliberately slow degree growth.

Expected Outcomes

Students should produce explicit ANF expressions (symbolic or algorithmic), plots or tables of degree growth, and a mathematical discussion explaining the security and efficiency implications.

2 Track 2: Linear Layer Theory and Diffusion

Objective

This track focuses on the linear diffusion layer of AO ciphers and its role in spreading information across the state.

Background

The linear layer of an AO cipher is typically a linear transformation:

$$L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n,$$

represented by an invertible matrix over $\mathbb{F}_2$.

Properties such as rank, branch number, and diffusion rate strongly influence resistance to differential, linear, and algebraic attacks.

Project Description

For a chosen AO cipher, explicitly construct the matrix corresponding to the linear layer. Analyze its algebraic and combinatorial properties, including:

- invertibility and rank,
- branch number or diffusion metrics,
- how linear mixing interacts with low-degree nonlinear layers.

Expected Outcomes

A rigorous linear-algebraic analysis of the diffusion layer, supported by proofs, examples, and clear explanations of how linear diffusion compensates for limited nonlinearity.

3 Track 3: Polynomial System Modeling of AO Encryption

Objective

This track studies AO ciphers as systems of polynomial equations over $\mathbb{F}_2$, laying the groundwork for algebraic cryptanalysis.

Background

Encryption under an AO cipher can be expressed as a system of multivariate polynomial equations:

$$F(x, k) = y,$$

where x denotes plaintext variables, k key variables, and y ciphertext variables.

The complexity of this system is influenced by degree, sparsity, and the number of equations.

Project Description

Construct the full polynomial system describing one or more rounds of an AO cipher. Analyze:

- the number of variables and equations,
- the maximum polynomial degree,
- structural sparsity and repeated monomials.

For very small instances, students may attempt solving the system using Gröbner basis techniques or SAT encodings, but the emphasis should remain on structural analysis rather than successful key recovery.

Expected Outcomes

A clear mathematical model of AO encryption as a polynomial system, accompanied by a discussion of why such systems are both attractive and dangerous from a security perspective.

Deliverables

Students must submit:

- A written report (8–12 pages) with full mathematical exposition
- Clear definitions, theorems, and proofs where appropriate
- Experimental or symbolic results supporting the analysis

Code is optional and could be used to support mathematical arguments.